

Cálculo Avançado

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1 Change of Variables

Theorem 1.1 (Existence of partitions of unity). *Let $\mathcal{A}$ be a collection of open sets in $\mathbb{R}^n$; let A be their union. There exists a partition of the unity $\{\phi_i\}_{i \in I}$ of A such that each ϕ_i is compactly supported in some $U \in \mathcal{A}$.*

Theorem 1.2. *Let A be open in $\mathbb{R}^n$, and let $f : A \rightarrow \mathbb{R}$ be continuous. Let $\{\phi_i\}$ be a partition of unity on A with compact supports. The integral $\int_A f$ exists if and only if the series*

$$\sum_{i=1}^{\infty} \left[\int_A \phi_i f \right]$$

converges. In this case,

$$\int_A f = \sum_{i=1}^{\infty} \left[\int_A \phi_i f \right].$$

Theorem 1.3 (Substitution rule). *Let $I = [a, b]$, and let $g : I \rightarrow \mathbb{R}$ be a C^1 function, with $g'(x) \neq 0$ for $x \in (a, b)$. Then $J = g(I)$ is a closed interval with endpoints $g(a), g(b)$. If $f : J \rightarrow \mathbb{R}$ is continuous, then*

$$\int_J f = \int_I (f \circ g) g'.$$

Definition 1.4 (Change of variables). *Let A be open in $\mathbb{R}^n$, and let $g : A \rightarrow \mathbb{R}^n$ be a one-to-one C^r function such that $\det Dg(x) \neq 0$ for $x \in A$. Then g is called a change of variables in $\mathbb{R}^n$.*

Theorem 1.5 (Change of variables theorem). *Let $g : A \rightarrow B$ be a diffeomorphism of open sets in $\mathbb{R}^n$, and let $f : B \rightarrow \mathbb{R}$ be a continuous function. Then f is integrable over B iff the function $(f \circ g) |\det Dg|$ is integrable over A . In this case,*

$$\int_B f = \int_A (f \circ g) |\det Dg|.$$

2 Diffeomorphisms in $\mathbb{R}^n$

Lemma 2.1. *Let A be open in $\mathbb{R}^n$, and let $g : A \rightarrow \mathbb{R}^n$ be a C^1 function. If the subset E of A has measure zero, then the set $g(E)$ also has measure zero.*

Theorem 2.2. *Let $g : A \rightarrow B$ be a C^r -diffeomorphism between open sets in $\mathbb{R}^n$, and let D be a compact subset of A and $E = g(D)$. Then we have:*

(a) $g(\text{int}D) = \text{int}E$ and $g(\partial D) = \partial E$.

(b) If D is rectifiable, so is E .

Definition 2.3 (Primitive diffeomorphism). Let $h : A \rightarrow B$ be a diffeomorphism of open sets in $\mathbb{R}^n$. If h preserves the i coordinate for some i , then h is called a primitive diffeomorphism.

Theorem 2.4. Let $g : A \rightarrow B$ be a diffeomorphism of open sets in $\mathbb{R}^n$, $n \geq 2$. Given $a \in A$, there is a neighbourhood $U_0 \subseteq A$ of a , and a sequence of diffeomorphisms

$$U_0 \xrightarrow{h_1} U_1 \longrightarrow \dots \xrightarrow{h_k} U_k,$$

such that the composite $h_k \circ \dots \circ h_2 \circ h_1$ equals $g|_{U_0}$ and such that each h_i is a primitive diffeomorphism.

Theorem 2.5. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation $h(x) = Hx$, where H is a matrix. Let S be a rectifiable set. Then

$$\text{vol}(h(S)) = |\det H| \text{vol}(S).$$

Definition 2.6 (Parellelopiped). Let $a_1, a_2, \dots, a_k$ be independent vectors in $\mathbb{R}^n$. We define $P = P(a_1, a_2, \dots, a_k)$ to be the set

$$P = \{c_1 a_1 + c_2 a_2 + \dots + c_k a_k : 0 \leq c_i \leq 1\}.$$

The a_i are called the edges of P .

Theorem 2.7. Let $a_1, \dots, a_n$ be independent vectors in $\mathbb{R}^n$. Let $A = [a_1 \dots a_n]$ be the n by n matrix with columns $a_1, \dots, a_n$. Then

$$\text{vol}(P(a_1, \dots, a_n)) = |\det A|.$$

Definition 2.8 (Orientation). Let V be an n dimensional vector space. An n tuple $(a_1, \dots, a_n)$ of independent vectors in V is called an n frame in V . In $\mathbb{R}^n$, we call such a frame right-handed if

$$\det[a_1, \dots, a_n] > 0.$$

We call it left-handed otherwise. The collection of all right-handed (or left-handed) frames in $\mathbb{R}^n$ is called an orientation. More generally, choose a linear isomorphism $T : \mathbb{R}^n \rightarrow V$ and define the orientations of V to be the images of the orientations of $\mathbb{R}^n$ via T .

Theorem 2.9. Let C be a non-singular matrix, and let h be the linear transformation associated to it. Let $(a_1, \dots, a_k)$ be a frame of $\mathbb{R}^n$. If $\det C > 0$, then the frames $(a_1, \dots, a_k)$ and $(h(a_1), \dots, h(a_k))$ belong to the same orientation. Else, they belong to opposite orientations.

Proposition 2.10. The orthogonal matrices form a group.

Proposition 2.11. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a map such that $h(0) = 0$. The following are equivalent:

- (i) h is an isometry,
- (ii) h preserves dot products, and

(iii) h is an orthogonal transformation.

Proposition 2.12. *Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Then h is an isometry iff it is the composition of a translation and an orthogonal transformation, that is*

$$h(x) = Ax + p,$$

where A is an orthogonal matrix.

Proposition 2.13. *Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an isometry. If S is rectifiable, then $h(S)$ is rectifiable and moreover $\text{vol}(h(S)) = \text{vol}(S)$.*

3 Manifolds

Lemma 3.1. *Let W be a k -dimensional linear subspace of $\mathbb{R}^n$. There is an orthogonal transformation $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that carries W onto the subspace $\mathbb{R}^k \times \{0\}$ of $\mathbb{R}^n$.*

Proposition 3.2. *There is a unique function V that assigns to each k -tuple $(x_1, \dots, x_k)$ of elements of $\mathbb{R}^n$, a non-negative number such that:*

(1) *If $h : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an orthogonal transformation, then*

$$V(h(x_1), \dots, h(x_k)) = V(x_1, \dots, x_k).$$

(2) *If $y_1, \dots, y_k$ belong to the subspace $\mathbb{R}^k \times \{0\}$ of $\mathbb{R}^n$, so that*

$$y_i = \begin{pmatrix} z_i \\ 0 \end{pmatrix}$$

for $z_i \in \mathbb{R}^k$, then

$$V(y_1, \dots, y_k) = |\det[z_1 \cdots z_k]|.$$

The function V vanishes iff the vectors x_i are linearly dependent. It satisfies the equation

$$V(x_1, \dots, x_k) = \left(\det(X^T X) \right)^{\frac{1}{2}},$$

where X is the n by k matrix $X = [x_1 \cdots x_k]$.

Definition 3.3 (Volume of parallelepiped). *If $x_1, \dots, x_k$ are independent vectors in $\mathbb{R}^n$, define the k -dimensional volume of $P = P(x_1, \dots, x_k)$ to be $V(x_1, \dots, x_k)$.*

Definition 3.4 (Parametrized manifold). *Let $k \leq n$. Let A be open in $\mathbb{R}^k$ and $\alpha : A \rightarrow \mathbb{R}^n$ be a C^k map. Then $M = \alpha(A)$ is a k -dimensional parametrized manifold. Define the (k -dimensional) volume of M to be*

$$\text{vol}(M) = \int_A V(D\alpha),$$

provided the integral exists.

Definition 3.5 (Integral of scalar function over M). Let $M = \alpha(A)$ be a k -dimensional parametrized manifold of class C^r . Let f be a real valued continuous function defined on M . Define the integral of f over M to be

$$\int_M f = \int_A (f \circ \alpha) V(D\alpha),$$

provided this integral exists.

Theorem 3.6 (Invariance under parametrization). Let $g : A \rightarrow B$ be a diffeomorphism of open sets in $\mathbb{R}^k$, and let $\beta : B \rightarrow \mathbb{R}^n$ be a C^r map. Let $M = \beta(B)$ and $\alpha = \beta \circ g$, then $\alpha : A \rightarrow \mathbb{R}^n$ and $\beta(B) = \alpha(A) = M$. If $f : M \rightarrow \mathbb{R}$ is a continuous function, f is integrable over $\alpha(A)$ iff it is integrable over $\beta(B)$, and in that case the integral coincides.

Definition 3.7 (k -manifold without boundary). Let M be a subspace of $\mathbb{R}^n$. We call M a k -manifold without boundary in $\mathbb{R}^n$ of class C^r if, for each $p \in M$, there are a set $V \ni p$ open in M , a set U open in $\mathbb{R}^k$, and a continuous map $\alpha : U \rightarrow V$, such that:

- (i) α is of class C^r .
- (ii) $\alpha^{-1} : V \rightarrow U$ is continuous.
- (iii) $D\alpha(x)$ has rank k for each $x \in U$.

The map α is called a coordinate patch.

Definition 3.8. Let S be a subset of $\mathbb{R}^k$, and let $f : S \rightarrow \mathbb{R}^n$. We say that f is of class C^r on S if it may be extended to a function $g : U \rightarrow \mathbb{R}^n$ that is of class C^r on an open subset $U \subseteq S$ of $\mathbb{R}^k$.

Lemma 3.9. Let $S \subseteq \mathbb{R}^k$, and let $f : S \rightarrow \mathbb{R}^n$. If for each $x \in S$, there is a neighbourhood U_x of x and a function $g_x : U_x \rightarrow \mathbb{R}^n$ of class C^r that agrees with f on $U_x \cap S$, then f is of class C^r on S .

Definition 3.10 (Upper half space). Let H^k denote the upper half space of $\mathbb{R}^k$, consisting of those $x \in \mathbb{R}^k$ for which $x_k \geq 0$. Let H_+^k denote the upper half space of $\mathbb{R}^k$, consisting of those $x \in \mathbb{R}^k$ for which $x_k > 0$.

Lemma 3.11. Let U be open in H^k but not in $\mathbb{R}^k$, and let $\alpha : U \rightarrow \mathbb{R}^n$ be of class C^r . Let $\beta : U' \rightarrow \mathbb{R}^n$ be a C^r extension of α defined on an open set $U' \subseteq \mathbb{R}^k$. Then for $x \in U$, the derivative $D\beta(x)$ depends only on the function α and is independent of the extension β . It follows that we may denote this derivative by $D\alpha(x)$ without ambiguity.

Definition 3.12 (k -manifold with boundary). Simply take the definition of manifolds without boundary and let the coordinate patches be defined on sets open on H^k instead of $\mathbb{R}^k$.

Lemma 3.13. The restriction of a coordinate patch (U, α) to an open set $V \subseteq U$ is again a coordinate patch.

Proposition 3.14. The transition functions of coordinate maps on overlaps are C^r if the coordinate maps are C^r , and their derivatives are non-singular.

Definition 3.15 (Boundary of a manifold). Let M be a k -manifold in $\mathbb{R}^n$ and let $p \in M$. If there is a coordinate patch $\alpha : U \rightarrow V$ on M about p such that U is open in $\mathbb{R}^k$, we say p is an interior point. Otherwise (if the coordinate patch must always be in H^k), then p is a boundary point, $p \in \partial M$.

Proposition 3.16 (Identifying boundary points). *Let M be a k -manifold in $\mathbb{R}^n$, and let $\alpha : U \rightarrow V$ be a coordinate patch about $p \in M$.*

- (a) *If U is open in $\mathbb{R}^k$, then p is an interior point of M .*
- (b) *If U is open in H^k and if $p = \alpha(x_0)$ for $x_0 \in H_+^k$, then p is an interior point of M .*
- (c) *If U is open in H^k and $p = \alpha(x_0)$ for $x_0 \in \mathbb{R}^{k-1} \times \{0\}$, then p is a boundary point of M .*

Theorem 3.17. *Let M be a k -manifold in $\mathbb{R}^n$ of class C^r . If ∂M is non-empty, then ∂M is a $k - 1$ manifold without boundary in $\mathbb{R}^n$ of class C^r .*

Proposition 3.18. *Let O be open in $\mathbb{R}^n$, and let $f : O \rightarrow \mathbb{R}$ be of class C^r . Let $M = f^{-1}(0)$ and let $N = \{x \in \mathbb{R}^n : f(x) \geq 0\}$. Suppose M is non-empty and $Df(x)$ has rank 1 at every point of M . Then N is an n -manifold in $\mathbb{R}^n$ and $\partial N = M$.*

Corollary 3.19. *The n -ball $B^n(a)$ is an n -manifold in $\mathbb{R}^n$ of class C^∞ and $S^{n-1}(a) = \partial B^n(a)$.*

Lemma 3.20. *A compact k -manifold admits a compactly supported finite partition of unity dominated by a given collection of covering coordinate patches.*

Definition 3.21. Let M be a manifold, $f : M \rightarrow \mathbb{R}$ continuous. Choose a partition of unity $\phi_1, \dots, \phi_l$ on M and define the integral of f over M as

$$\int_M f = \sum_{i=1}^l \left[\int_M (\phi_i f) \right].$$

Proposition 3.22. *The integral is $C^\infty(M)$ linear.*

Definition 3.23. A subset $D \subseteq M$ of a manifold M is said to have measure zero in M if there are countably many coordinate patches such that the preimages of D via these maps have measure zero in $\mathbb{R}^k$.

Proposition 3.24. *Let M be a manifold, $f : M \rightarrow \mathbb{R}$ a continuous function. If M can be written as the union of finitely many parametrized manifolds $\alpha_i : A_i \rightarrow M_i$ and a set of measure zero in M , then the integral of f on M can be evaluated by summing the contributions on each parametrized piece.*

4 Differential Forms

Tensors and their properties are defined and proved as usual, alternating tensors, dual of a linear transformation, and differential forms as well.

Proposition 4.1. *Let A be open in $\mathbb{R}^k$ and $\alpha : A \rightarrow \mathbb{R}^n$ be a C^∞ function. Let $x \in \mathbb{R}^k$ and $y \in \mathbb{R}^n$. Then*

- (a) $\alpha^*(dy_i) = d\alpha_i$.
- (b) *If $I = (i_1, \dots, i_k)$ is an ascending k -tuple from the set $\{1, \dots, n\}$, then*

$$\alpha^*(dy_I) = \left(\det \frac{\partial \alpha_I}{\partial x} \right) dx_1 \wedge \dots \wedge dx_k.$$

5 Stokes' Theorem

Definition 5.1. Let $A \subseteq \mathbb{R}^k$ be an open set, and let $\eta \in \Omega^k(A)$. Then if $\eta = f dx_1 \wedge \cdots \wedge dx_k$, we define the integral of η as

$$\int_A \eta = \int_A f.$$

Definition 5.2. Let $A \subseteq \mathbb{R}^k$ be an open set and $\alpha : A \rightarrow \mathbb{R}^n$ be a C^∞ map. Then if $M_\alpha = \alpha(A)$ is a parametrized manifold and ω is a k -form defined in an open set of $\mathbb{R}^n$ containing M , we define

$$\int_M \omega = \int_A \alpha^* \omega,$$

provided the integral exists. The integral is linear.

Proposition 5.3. *The integral is invariant under reparametrization up to a sign.*

Definition 5.4 (Orientation preserving map). Let $g : A \rightarrow B$ be a diffeo of open sets in $\mathbb{R}^k$. We say that g is orientation preserving if $\det Dg > 0$ on A . If $\det Dg < 0$, then g is orientation reversing.

Definition 5.5 (Orientable manifold). Let M be a k -manifold in $\mathbb{R}^n$. If M has an atlas on which the transition functions are orientation preserving, then we say M is orientable. Otherwise, we say M is non-orientable.

Definition 5.6 (Oriented manifold). Let M be an orientable manifold. Given a collection of coordinate patches which overlap positively (i.e., the transition function is orientation preserving), take the maximal atlas on M which contains all coordinate patches which overlap positively with the given ones. Call this an orientation on M . Then M with an orientation is called an oriented manifold.

Proposition 5.7 (On 1-manifolds). *In the case of a 1-manifold, the allowed domains of the coordinate patches are open sets in $\mathbb{R}^1$, H^1 , or L^1 , the left half-line. With this convention, every 1-manifold is orientable.*

Definition 5.8 (Unit normal to a $(n-1)$ -manifold). Let M be an $n-1$ manifold in $\mathbb{R}^n$. Let n be a unit vector on $T_p \mathbb{R}^n$ that is orthogonal to $T_p M$. Then n is uniquely determined up to a sign. Given an orientation on M , choose a coordinate patch α about p belonging to the orientation. Then the columns $\frac{\partial \alpha}{\partial x_i}$ of the matrix $D\alpha$ give a basis for $T_p M$, and we specify the sign of n by requesting that

$$[n D\alpha(x)]$$

have positive determinant at $\alpha(x) = p$.

Definition 5.9 (Natural orientation of n -manifold). Let M be an n -manifold in $\mathbb{R}^n$. We define the natural orientation on M to consist of all the coordinate patches α with $\det D\alpha > 0$.

Definition 5.10. Let M be an oriented k manifold. The reflection map $r : \mathbb{R}^k \rightarrow \mathbb{R}^k$,

$$r(x_1, x_2, \dots, x_k) = (-x_1, x_2, \dots, x_k),$$

which takes H^k to H^k if $k > 1$ and H^1 to L^1 , reverses the orientation on M by acting on the coordinate patches α_i ,

$$\beta_i = \alpha_i \circ r.$$

Proposition 5.11. *Let $k > 1$. If M is an orientable k -manifold with non-empty boundary, then ∂M is orientable.*

Definition 5.12 (Induced orientation on ∂M). Let M be an orientable k -manifold with non-empty boundary ∂M . Given an orientation on M , the induced orientation on ∂M will be as follows: if k is even, it is the orientation obtained by restricting the coordinate patches belonging to the orientation on M to the boundary; if k is odd, it is the reverse orientation.

Definition 5.13. Redefine the integral of a k -form over a compact oriented k -manifold in $\mathbb{R}^n$ in the same way as before, with M being a union of parametrized manifolds and a set of measure zero on M , but use coordinate patches belonging to the orientation on M , and so the integral will be

Theorem 5.14 (Stokes' Theorem). *Let $k > 1$. Let M be a compact oriented k -manifold in $\mathbb{R}^n$, and give ∂M the induced orientation of it is not empty. Let ω be a $(k-1)$ -form defined in an open set of $\mathbb{R}^n$ containing M . Then*

$$\int_M d\omega = \int_{\partial M} \omega$$

if ∂M is not empty, and $\int_M d\omega = 0$ if ∂M is empty.